

Pass/Fail Check Pipeline

Objective

Design and implement a basic data pipeline solution to process historical stock market data, specifically focusing on the US indexes. Flag significant price movements based on configurable thresholds. A user with access to the pipeline code should be able to run the pipeline and produce the results as a file (e.g. csv)

Requirements

1. **Data Source(s):**
 - Use a dataset for the stock prices of market indexes provided as csv files
2. **Check Logic:**
 - Check 1: Flag day-over-day changes that are greater than 1%
 - Check 2: Flag week-over-week changes that are greater than 5%
 - Allow adding customizable thresholds for specific indexes.
Example: A user should be able to configure the pipeline to only flag daily movement of SP500 if it exceeds 1.5%, while the rest continue to flag the 1% changes
3. **Results:**
 - A csv file with a list of the items that break the thresholds set in the above check logic. The minimum set of information to include in the results
 - Values being compared
 - Difference in values
 - Ticker symbols associated with each break
 - Any other fields that would help a downstream system or end user interpret the results
4. **Interface:**
 - Running the pipeline can be done from the command line as long as it is clearly documented and the pipeline output is clear for viewing. A UI is not required.
 - Configuring the metrics for a pipeline run should be done via database records or a static file (json or yaml). The configuration should govern the following:
 - Check configuration (e.g. threshold)
 - On/Off switches for the checks

Deliverables

- A working pipeline that ingests data, applies the rules and outputs results.
- Source code (publically-access github repo or a compressed file with the source code).
- README with any setup steps and run instructions.

Expectations

- Submit the solution 24 hours before the interview in a GitHub repo.
- While the provided data set is small, the final solution should scale to large data sets (10s of thousands of rows)
- If applicable, state how/where a coding assistant is used, preferably in code comments. Commit code at intermediate stages, particularly when code is generated.
- Explain and modify the code during the interview call without the use of AI assistance.
- Discuss design choices made as well as the choice of any libraries used in the code
- Discuss the use of the static file-based configuration vs. the use of the pipeline storage. E.g. what kind of configuration is better fit for each